

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT1 — CHART CONSTRUCTION TEST

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Construction (BL3)	Assessment:	Assessment Tool 1 – Chart Construction
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Apply appropriate chart construction techniques — correct chart type, axes, scales, and calculations — to accurately represent given data.(BL3)</i>		
Student Name:	RAGUL.M	Reg. No.:	192524066
Section / Batch:	SLOT A	Date:	25/07/2026

Code of Conduct

I RAGUL.M(1925224066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., percentages, angles, bin ranges, five-number summaries) as part of your answer.
- Every chart must include a title, correctly labeled axes, and a legend where applicable.

Annexure A – CHART CONSTRUCTION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Chart Construction Test

CO2 · BT3: Apply ·

This rubric is used to assess student responses to the CO2-AT1 Chart Construction Test, in which students are given a small dataset and must construct an accurate, correctly labeled chart of a specified type.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type, and the axes, scale, and elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type, axes, or scale.	Poor understanding of core construction concepts; selects a chart type or elements fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., angles, bin ranges, five-number summary) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	15%	The chart is neatly and logically constructed, with a title, correctly labeled axes, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, axes, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, axis labels, or a legend.
Accuracy	10%	The chart is completely	Mostly accurate, with only minor	Partially correct; some plotted	Largely inaccurate; major

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		accurate — all values, proportions, scales, and calculations correctly match the given data.	omissions or a small error in one value or calculation.	values, proportions, or calculations are missing or incorrect.	errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 21 – Choropleth-Style Grid Map

Title

Choropleth-Style Grid Map Using Sequential Color Scale

Objective

To visualize the sales index of different regions using a sequential color scale, where lighter shades represent lower values and darker shades represent higher values.

Procedure

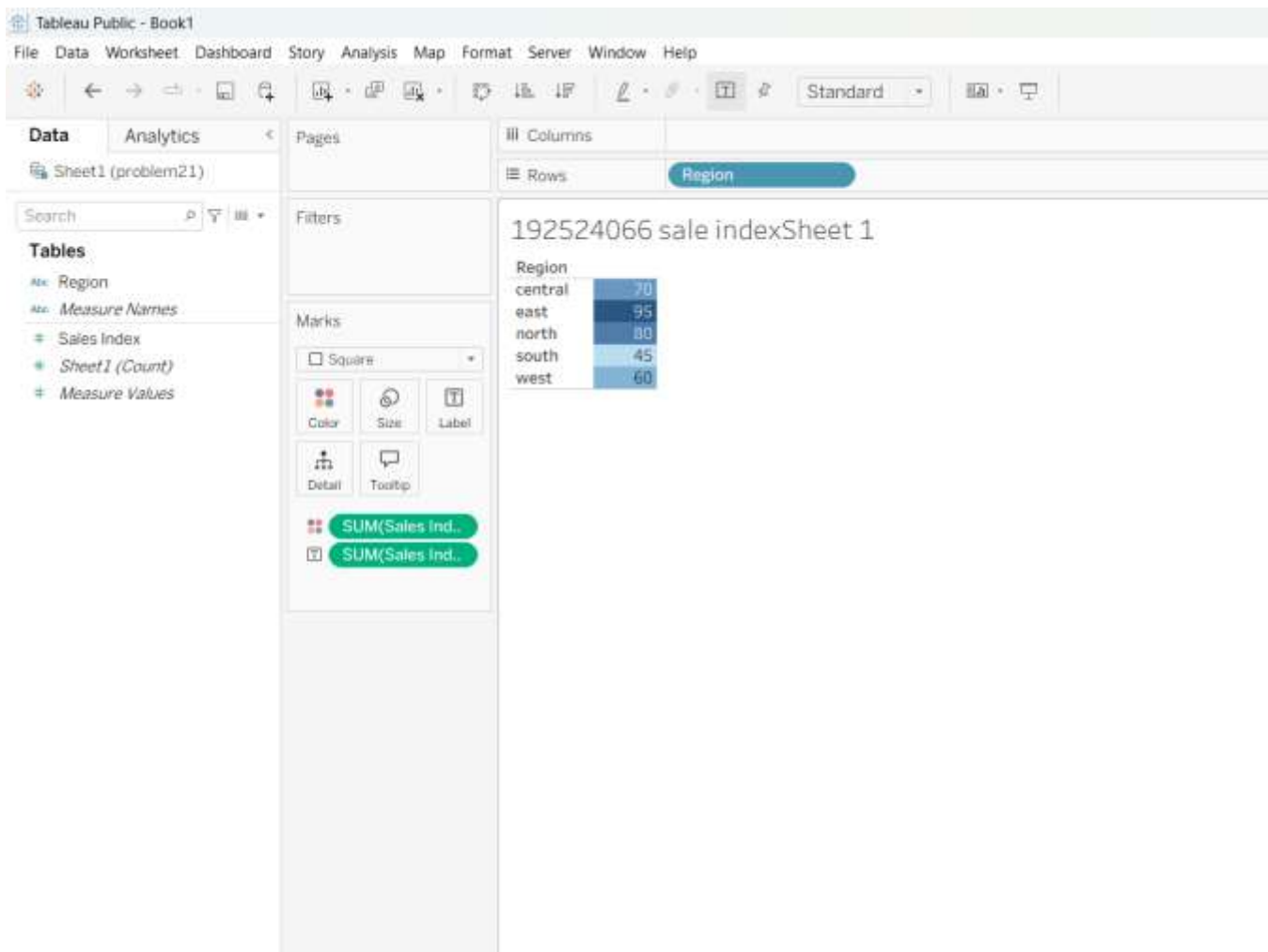
- Imported the sales index dataset into Tableau.
- Placed the Region field in the view.
- Applied the Sales Index field to the Color shelf.
- Selected a Sequential Color Palette.
- Generated a grid-style map where each region is shaded according to its sales index.

Observation

- East (95) appears as the darkest region.
- South (45) appears as the lightest region.
- Other regions are shaded according to their sales index values.

Result

A choropleth-style grid map was successfully created to represent regional sales intensity using a sequential color scale.



Problem 22 – Diverging Color Scale (Profit/Loss)

Title

Bar Chart Using Diverging Color Scale for Profit and Loss

Objective

To visualize monthly profit and loss values using different colors for positive and negative values while highlighting the zero baseline.

Procedure

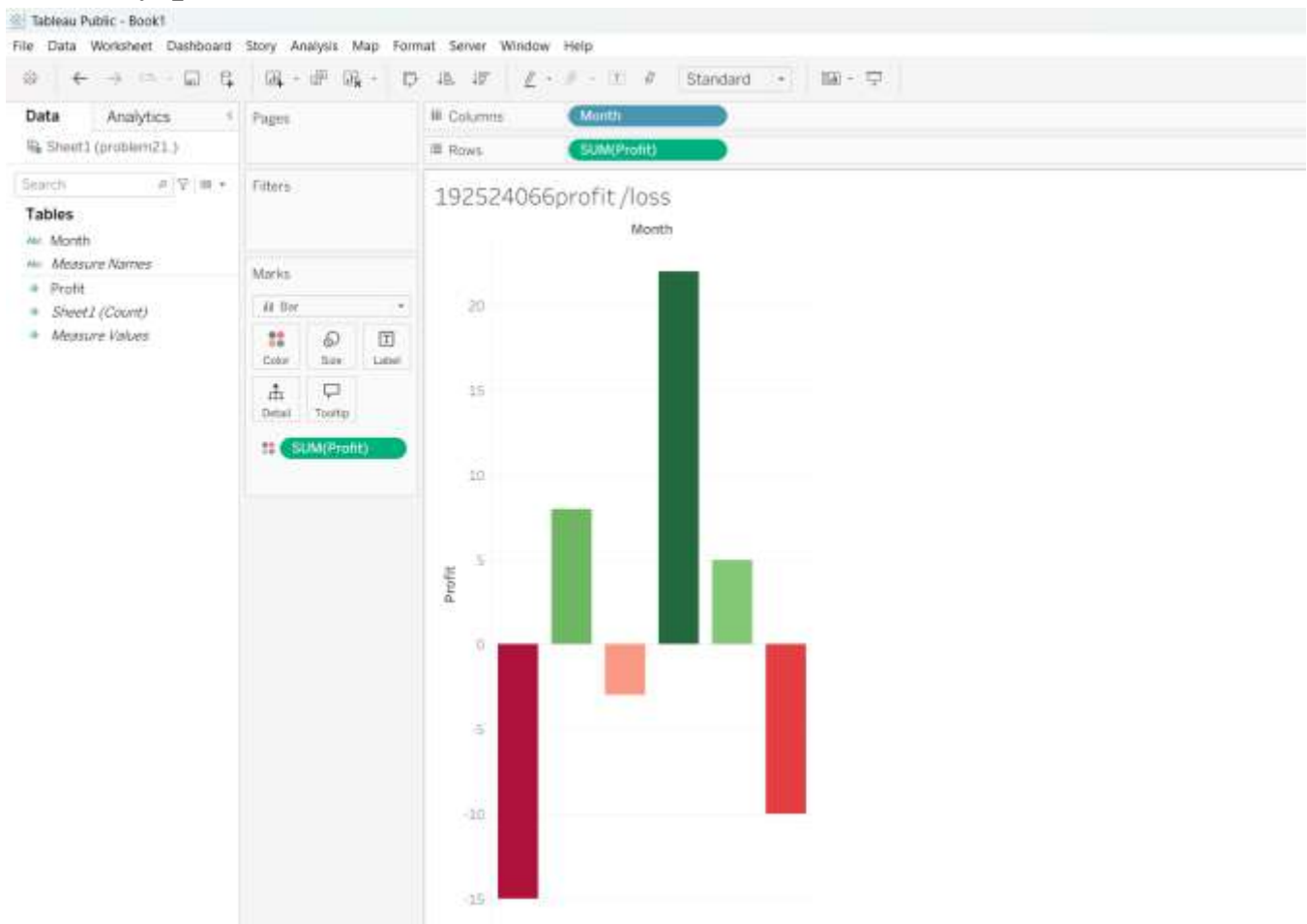
- Imported the monthly profit/loss dataset into Tableau.
- Created a Bar Chart using Month and Profit/Loss.
- Applied a Diverging Color Palette.
- Set the color midpoint to Zero.
- Displayed the zero baseline for easy comparison.

Observation

- Positive profits are displayed in one color.
- Negative profits are displayed in another color.
- The zero baseline clearly separates profits from losses.

Result

A diverging color bar chart was successfully created to distinguish monthly profits and losses.



Problem 23 – Treemap

Title

Treemap for Product Category Revenue

Objective

To represent revenue distribution among product categories using proportional rectangle sizes.

Percentage Calculation

Category Revenue Percentage

Electronics 400K 40%

Clothing 250K 25%

Books 150K 15%

Toys 100K 10%

Home 100K 10%

Total Revenue = 1000K

Procedure

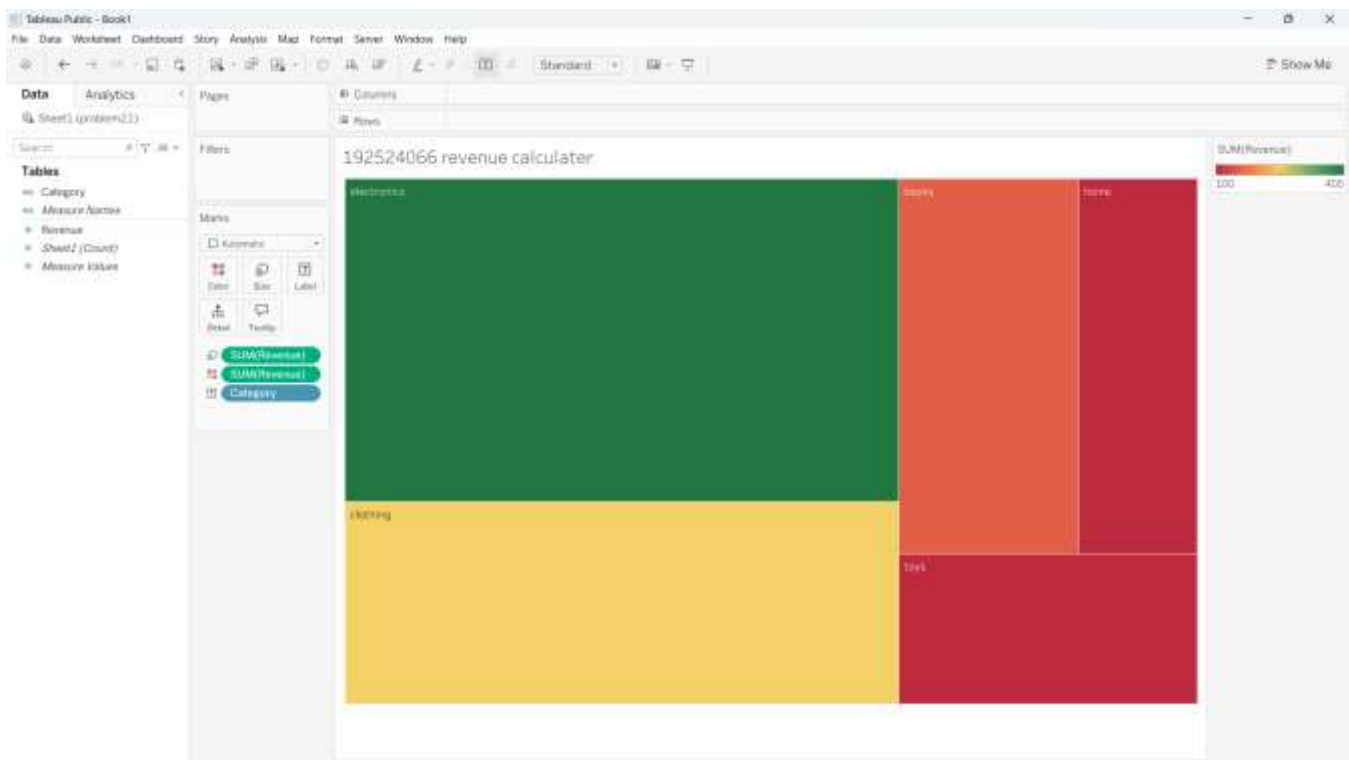
- Imported the revenue dataset into Tableau.
- Selected the Treemap visualization.
- Assigned Category to labels.
- Assigned Revenue to size and color.
- Generated the treemap based on proportional revenue values.

Observation

- Electronics occupies the largest area.
- Clothing occupies the second largest area.
- Toys and Home have equal-sized rectangles.
- Rectangle size reflects each category's revenue share.

Result

A treemap was successfully created to visualize proportional revenue distribution across product categories.



Problem 24 – Combination Chart (Bar + Line)

Title

Combination Chart Showing Monthly Sales and Average Price

Objective

To compare monthly sales units and average price using bars and a line chart.

Procedure

- Imported the monthly sales dataset into Tableau.
- Created a Bar Chart for units sold.
- Added a Line Chart for average price.
- Used different visual styles to distinguish the two measures.
- Displayed both variables across the six months.

Observation

- Sales units vary from month to month.
- Average price also changes over time.
- The chart enables comparison of sales volume and pricing trends.

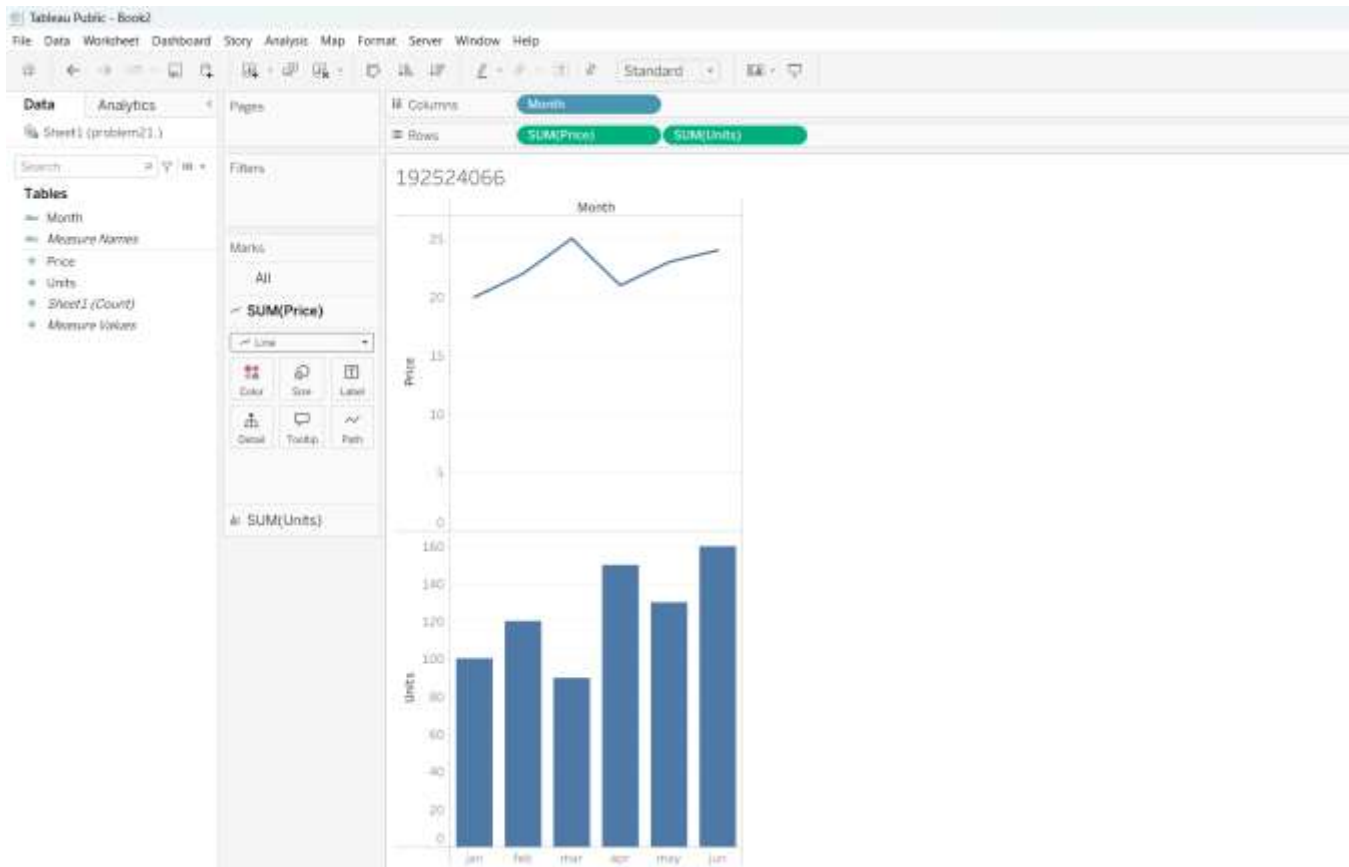
Why Avoid a Dual Y-Axis?

- Dual axes can be misleading because the variables use different units.

- They may suggest relationships that do not actually exist.
- A clearer approach is to normalize the data or use separate aligned charts sharing the same time axis.

Result

A combination chart was successfully created to compare monthly sales and average price while maintaining clear data interpretation.



Problem 25 – Small Multiples Construction

Title

Small Multiple Line Charts for Regional Weekly Sales

Objective

To compare weekly sales trends across four regions using separate line charts.

Procedure

- Imported the weekly sales dataset into Tableau.
- Created a Line Chart using Week and Sales.
- Split the visualization by Region.
- Generated four separate line charts with a common layout.
- Ensured each chart used the same scale for comparison.

Observation

- Region A shows a steady increase in sales.
- Region B exhibits moderate fluctuations.
- Region C records the highest sales throughout the period.
- Region D has the lowest sales but shows gradual improvement.

Justification

Small multiples reduce visual clutter and overlapping lines, making it easier to compare trends across regions while maintaining a consistent scale.

Result

A set of four small multiple line charts was successfully created, providing a clear comparison of weekly sales trends for all regions.

This content is ready to paste into your assessment document alongside your Tableau screenshots.

